



DPUCANSensoWheel

The DPUCANSensoWheel provides an interface to the SENSO-Wheel drive system manufactured by SENSODRIVE GmbH. This DPU operates the SENSO-Wheel in the Normal Mode, and (depending on the configuration) modifies the parameters Stiffness, Damping and Friction according to the vehicle speed.

This manual is based on the manuals "Software Manual SENSO-Wheel version 3.7.0" and "Operating Manual SENSO-Wheel version 5.0.1" by SENSODRIVE GmbH, Argelsrieder Feld 20 TE04 in D-82234 Weßling.

1. Inputs:

<i>v</i>	Current vehicle speed [km/h]
<i>DOut[1...3]</i>	State of the digital outputs [0/1]
<i>ControlPosition</i>	If the SENSO Wheel firmware allows the position control mode, the desired position [°] will be set with this value. Default 0°

2. Dynamic parameters:

<i>Mode</i>	Operating mode of the SENSO Wheel. The individual operating modes are explained in more detail in the section "Functional principle".
<i>EnableCANWatchdog</i>	Flag which activates the CAN watchdog [0/1]. Default: 0
<i>EnableFriction2</i>	Flag which activates the friction version 2 [0/1]. Default: 1
<i>EnableVibration</i>	Flag which activates vibration when the end stops are reached [0/1]. Default: 0
<i>EnableWheelPositionControl</i>	Flag which activates the position control mode. Default 0.
<i>GainKsw</i>	Gain [0...200] for SENSO Wheel ksw (see SENSO Wheel manual). Default 100.
<i>GainKlw</i>	Gain [0...200] for SENSO Wheel klw for position control mode (see SENSO Wheel manual). Default 0
<i>i_SW_ControlPosition</i>	Factor for input ControlPosition. Default 1.0. Use 57.2958 for ControlPosition in rad.
<i>i_SW_WheelReferencing</i>	Factor for output WheelPositionXXX. Default 1.0 Configures referencing mode: 0 = No referencing 1 = Referencing to index signal 2 = Referencing to mechanical end stops Default: 0
<i>Ref2Damping</i>	Damping for the referencing 2 [mNm/1/min]. Default: 200
<i>Ref2Torque</i>	Torque for the referencing mode 2 [mNm]. Default: -2000
<i>Ref2Pos</i>	Position of the end stops [°]. Default: 738



<i>Ref2Steps</i>	Time steps for detection of the end stops [-]. Default: 50
<i>WheelPositionOffset</i>	Offset for the position of the index signal in referencing mode 1 [Grad]. Range: -359° up to +359°. Default: 0°
<i>WheelPositionMax</i>	Position of the end stops of the controller [°]. Range: 10° - 1440°. Default: 360°
<i>Torque</i>	Desired torque setting for Normal mode [mNm]. Range -32000 ... +32000. Default: 0
<i>Friction</i>	Friction in Normal mode [mNm]. Range 0 ... 5000. De-fault: 0
<i>Damping</i>	Damping in Normal mode [mNm/1/min]. Range 0 ... 500. Default: 0
<i>Stiffness</i>	Spring stiffness in Normal mode [mNm/1°]. Range 0 ... 2500. De-fault: 0
<i>GainMu, GainFD, GainWS</i>	Gain for the parameters Torque, Friction, Damping and Stiffness (see section "Functional principle").
<i>FrictionGain</i>	Gain for the parameter Friction. Default: 1.0
<i>DampingGain</i>	Gain for the parameter Damping. Default: 1.0
<i>StiffnessGain</i>	Gain for the parameter Stiffness. Default: 1.0

3. Optional parameters of the SENSO-Wheel controller:

The optional parameters of the SENSO-Wheel Effects (which are explained in section 7.3 of the software manual) are listed in this section.

<i>OptEndStiffness</i>	Stiffness of end stops [mNm/1°]. Default: 2500
<i>OptEndDamping</i>	Damping at end stops [mNm/1/min]. Default: 300
<i>OptRefDamping</i>	Damping during the referencing mode 1 [mNm/1/min]. Default: 200
<i>OptFriGain</i>	Gain of slope for friction [-]. Default: 5000
<i>OptFriLimit</i>	Friction limit [mNm]. Default: 1500
<i>OptVEndDist</i>	Distance of vibration beyond the end stops [°]. Default: 50
<i>OptVEndPeriod</i>	Period of vibrations beyond the end stops [ms]. Default: 100
<i>OptVEndTime</i>	Time without torque beyond the end stops [ms]. Default: 25
<i>OptFri2Damping</i>	Damping for Friction Version 2 [mNm/1/min]. Default: 400
<i>OptFri2Factor</i>	Factor for Friction Version 2 [%]. Default: 100
<i>OptDampingVhalf</i>	Damping: velocity at which the half damping value is reached [0.1 1/min]. Default: 0
<i>OptDampingKT1</i>	Damping: velocity filter constant kT1 for T1 = 65.536 / kT1 [-]. Default: 65535
<i>OptCurrentSwitch</i>	If set to 1 the next optional parameters will be set.
<i>OptCurrentKp</i>	Kp value [1/16] for current controller. Default 5.0.
<i>OptCurrentKi</i>	Ki value [4s/Tn] for current controller. Default 1300.0

4. Static parameters:

<i>CANMode</i>	Selects a CAN bus channel for communication, which is managed by DPUCANBUS (see also DPUCANBUS.pdf).
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5. Outputs:

<i>DIn[1...6]</i>	State of the digital inputs 1- 6
<i>Aln[1...2]</i>	Voltage at analog inputs 1 and 2
<i>WheelPosition</i>	Steering wheel position [°]
<i>WheelPositionRad</i>	Steering wheel position [rad]



<i>WheelPositionP</i>	Steering wheel velocity, computed from the steering wheel position [°/s]
<i>WheelVPosition</i>	Steering wheel velocity, measured by the sensor [1/min].
<i>WheelGain</i>	Resulting gain factor, according to the following equation: $WheelGain = Gain \cdot GainMu \cdot GainFD \cdot GainWS$
<i>WheelTorque</i>	Actual torque [mNm]
<i>WheelFriction</i>	Current friction value (at the controller) [mNm]
<i>WheelDamping</i>	Current damping value (at the controller) [mNm/°]
<i>WheelStiffness</i>	Current spring stiffness value (at the controller) [mNm/1/min]
<i>WheelState</i>	Controller status
<i>InkPerU</i>	Number of increments per revolution

6. Dependencies to other DPUs:

In addition to **DPUCANSensoWheel**, a **DPUCANBUS** with the same *CANMode* must be running on the same computer, which is connected to the CAN hardware. The *Index* value of **DPUCANBUS** must be lower than the *Index* value of **DPUCANSensoWheel**.

7. Functional principle:

The SENSO-Wheel is always operated in the Normal Mode (see chapter 6.3 in the SENSO-Wheel software manual). The Basic Mode (chapter 6.4 of the SENSO-Wheel software manual) is not supported.

In the Normal Mode, the SENSO-Wheel is controlled by the parameters Friction, Stiffness und Damping. Depending on the Mode parameter, these parameters are modified by lookup tables that depend on the vehicle speed.

At first, an internal gain (*GainSW*) of the parameters is computed independent of the operating mode. The inputs *GainMu*, *GainFD* and *GainWS* are intended to influence the steering moment from different sources (e.g. the road's friction coefficient, inputs from the vehicle dynamics and the weather). For example, the steering moment can be reduced drastically when the road is icy.

$$GainSW = Gain \cdot GainMu \cdot GainFD \cdot GainWS$$

Depending on the operation mode, the parameters *TorqueSW*, *FrictionSW*, *StiffnessSW* and *DampingSW* that are sent to the controller, are computed using the formulas de-tailed below.

Mode	Operation mode
0	Normal Mode
1	Normal Mode, parameters are modified using speed-dependent lookup tables
2	SENSODrive-internal lookup tables are used
3	SENSODrive lookup tables · speed-dependent lookup tables
4	Constant moment for spring stiffness



In Mode 0, the inputs *Friction*, *Stiffness*, *Damping* and *Torque* are multiplied by the corresponding Gain parameters and then sent to the controller.

$$TorqueSW = Torque \cdot GainSW$$

$$FrictionSW = Friction \cdot FrictionGain \cdot GainSW$$

$$StiffnessSW = Stiffness \cdot StiffnessGain \cdot GainSW$$

$$DampingSW = Damping \cdot DampingGain \cdot GainSW$$

In Mode 1, the inputs are additionally modified by the speed-dependent lookup tables.

$$TorqueSW = Torque \cdot GainSW$$

$$FrictionSW = (vFLUT(v) \cdot FrictionGain + Friction) \cdot GainSW$$

$$StiffnessSW = (vSLUT(v) \cdot StiffnessGain + Stiffness) \cdot GainSW$$

$$DampingSW = (vDLUT(v) \cdot DampingGain + Damping) \cdot GainSW$$

The lookup tables *vFLUT*, *vSLUT* and *vDLUT* can be defined in the SILAB configuration by inputting pairs of speed and coefficient values, like in the following example:

```
DPUCANSensoWheel ~Wheel
{
    Computer = {CAN};
    Index = 2;
    CANMode = 1;

    #####
    ## Mode 1: vFLUT / vSLUT / vDLUT = ( <vkm_h> , <F/S/D> [,...] );
    Mode = 1;

    vFLUT = (
        0.0 , 2160.0,
        5.0 , 0.0,
        200.0 , 0.0
    );
    vSLUT = (
        0.0 , 0.0,
        10.0 , 40.0,
        200.0 , 45.0
    );
    vDLUT = (
        0.0 , 0.0,
        5.0 , 36.0,
        200.0 , 36.0
    );
};
```



```
};

Connections =
{
    VDyn.v_kmh -> ~Wheel.v
};
```

The lookup tables are defined by sampling points that are defined by pairs of a speed and a coefficient value. Linear interpolation is used between the sampling points.

In the example above, the lookup table vFLUT achieves a high friction (value 2160) while the vehicle is standing. The friction vanishes when the vehicle drives faster than 5 km/h. The spring stiffness is missing completely while standing, but increases steeply up to 10 km/h and further slightly up to 200 km/h.

The lookup tables used in Mode 2 are derived from studies performed by SENSODRIVE GmbH and cannot be modified or read-out.

In Mode 2, the inputs Friction, Stiffness and Damping do not influence the steering moment at all. The outputs WheelFriction, WheelStiffness and WheelDamping are constantly set to 0.0.

$$TorqueSW = Torque \cdot GainSW$$

$$FrictionSW = FrictionSENSODRIVE(v) \cdot FrictionGain \cdot GainSW$$

$$StiffnessSW = StiffnessSENSODRIVE(v) \cdot StiffnessGain \cdot GainSW$$

$$DampingSW = DampingSENSODRIVE(v) \cdot DampingGain \cdot GainSW$$

In Mode 3, the lookup tables defined in Mode 2 are additionally multiplied by user-defined lookup tables vFLUT, vSLUT and vDLUT. The user-defined tables are defined like in Mode 1, however, in Mode 3, the second tuple entry is a multiplier for the SENSODRIVE lookup tables.

$$TorqueSW = Torque \cdot GainSW$$

$$FrictionSW = FrictionSENSODRIVE(v) \cdot vFLUT(v) \cdot FrictionGain \cdot GainSW$$

$$StiffnessSW = StiffnessSENSODRIVE(v) \cdot vSLUT(v) \cdot StiffnessGain \cdot GainSW$$

$$DampingSW = DampingSENSODRIVE(v) \cdot vDLUT(v) \cdot DampingGain \cdot GainSW$$

The following example explains user-defined lookup tables for Mode 3:

```
DPUCANSensoWheel ~Wheel
{
    Computer = {CAN};
    Index = 2;
    CANMode = 1;

    #####
    ## Mode 3: vFLUT / vSLUT / vDLUT = ( <vkm_h> , <xF/xS/xD> [,...] );
```



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version Version 7.1 documentation

```
Mode = 3;

vDLUT = ( 0.00,1.10,
          20.83,1.10,
          23.61,0.95,
          30.00,0.95);

vSLUT = ( 0.00,1.10,
          20.83,1.10,
          23.61,0.95,
          30.00,0.95);

};

Connections =
{
    VDyn.v_kmh -> ~Wheel.v
};
```

In Mode 4, the parameter *StiffnessSW* is modified depending on the steering angle. The moment will not rise any further as soon as the steering angle exceeds 1 rad. Below 1 rad, the parameters are computed just like in Mode 0.

$$TorqueSW = Torque \cdot GainSW$$

$$FrictionSW = Friction \cdot FrictionGain \cdot GainSW$$

$$DampingSW = Damping \cdot DampingGain \cdot GainSW$$

The spring stiffness is computed according to the following fomula:

$$StiffnessSW = Stiffness/\max(1,| WheelPositionRad |) \cdot StiffnessGain \cdot GainSW$$

8. Referencing:

The SENSO-Wheel drive system does not include an absolute position signal. To ensure correct centering of the steering wheel, referencing can be performed when starting up the system. If no referencing is performed, the position of the steering wheel after power-up is assigned to the center (0°) position.

The Referencing parameter controls if, and what kind of, referencing is performed.

Referencing	Effect
0	No referencing
1	Referencing to index signal
2	Referencing to mechanical end stops

If referencing to an index signal, the steering wheel rotates after the launch step until the index signal has been found. Usually, the index signal is found after at most one clockwise revolution of the steering wheel. Depending on the motor's mounting angle, the position of the index signal may differ from



the desired home position of the steering wheel. In that case, the home position can be corrected using the parameter *WheelPositionOffset*.

```
DPUCANSensoWheel ~Wheel
{
    Computer = {CAN};
    Index = 2;
    CANMode = 1;
    Referencing = 1;
    WheelPositionMax = 720;
    WheelPositionOffset = -12;
};
```

In the example above, the position had to be corrected by -12° . In this case, the steering wheel will first search for the index signal, and then move to the corrected home position. The referencing is performed again whenever the controller's power is switched on.

If the drive system is equipped with mechanical end stops, referencing mode 2 should be used. In this mode, the steering wheel first revolves in clockwise direction until the steering angle no longer changes for *Ref2Steps* simulation cycles. This position is assigned the value *Ref2Pos* and the steering wheel returns to the home position (which is now known).

The referencing mode 2 is only performed once after the steering wheel controller has been powered on.

```
DPUCANSensoWheel ~Wheel
{
    Computer = {CAN};
    Index = 2;
    CANMode = 1;
    Referencing = 2;
    WheelPositionMax = 720;
    Ref2Damping = 200;
    Ref2Torque = -2000;
    Ref2Pos = 738;
    Ref2Steps = 50;
};
```

In the example above, the end stop's position is 738° . In order to protect the end stop during the simulation, the controller's end stop is set to the lower value of 720° .